

Embedding Models and Vector Representations

Embedding models transform raw text, images, or other data into high-dimensional vectors that capture semantic meaning. Text embedding models like Sentence-BERT, MPNet, and OpenAI's text-embedding models encode documents and queries into vector spaces where semantically similar texts are positioned close together. The dimensionality of embeddings ranges typically from 384 to 1536 dimensions or higher, with increased dimensionality generally capturing more nuanced semantic information but requiring more storage and computation. Transformer-based models dominate embedding generation, leveraging attention mechanisms to understand context and relationships within text. Contrastive learning during model training ensures that similar texts map to nearby vector locations while dissimilar texts are far apart. Domain-specific embedding models fine-tuned on specialized corpora outperform general models in particular domains. Multi-lingual embedding models enable cross-language semantic search. Recent developments include instruction-following embeddings that allow specifying retrieval intent. Normalized embeddings ensure consistent similarity computations. Understanding embedding model characteristics—speed, accuracy, dimensionality, language support, and specialized capabilities—is essential for effective RAG system design.